

Size Polymorphisms in the Human Ultrahigh Sulfur Hair Keratin-Associated Protein 4, *KAP4*, Gene Family

Naoyuki Kariya, Yutaka Shimomura, and Masaaki Ito

Department of Dermatology, Niigata University School of Medicine, Niigata, Japan

Hair keratin-associated proteins (KAP) are a major structural component of the hair fiber. In humans, five clusters of *KAP* genes have been identified, leading to the isolation of more than 80 individual *KAP* members. Of these, a cluster of high/ultrahigh sulfur *KAP* has been characterized on chromosome 17q12–21. Within this cluster, however, there was an incompletely sequenced region in which the ultrahigh sulfur *KAP4* genes were located. A recently reported human bacterial artificial chromosome clone, AC100808, finally covered the gap, which enabled us to characterize the complete set of *KAP4* genes in this cluster. Analysis of the nucleotide sequences of AC100808, together with PCR amplification, allowed us to identify numerous size polymorphisms in the *KAP4* genes, which were mainly due to variations in the sequences encoding cysteine-rich repeat segments. Taken together, the data indicate that the polymorphic alleles of the *KAP4* genes may have arisen through intragenic deletion and/or duplication of sequences encoding the repeat structures during evolution.

Key words: cysteine-rich repeat segments/hair keratin-associated proteins/*KAP4* family/size polymorphisms
J Invest Dermatol 124:1111–1118, 2005

Morphologically, hair fiber is composed of an external cuticular sheath, an inner cortex, and in some cases, a central medulla. The formation of the hair fiber largely involves the expression of hair keratins and their associated proteins. The hair keratins form intermediate filaments (KIF) in trichocytes, which are cells that generate the central hair-forming compartment. In the cortex region, as the cortical cells move upward, the KIF gradually aggregate, and each KIF, about 8–10 nm in diameter, is surrounded by an amorphous space called the matrix. The hair keratin-associated proteins (KAP) are a major component of the matrix between the KIF, and are thought to form the rigid hair shaft through a cross-linked network with the KIF (Powell and Rogers, 1997).

KAP are encoded by a large number of multigene families. The *KAP* genes, in general, consist of a single exon of less than 1000 bp. To date, more than 100 *KAP* genes and/or proteins have been isolated from humans and other mammalian species. KAP are classified into three groups on the basis of their amino acid composition: high sulfur (<30 mol% cysteine content), ultrahigh sulfur (>30 mol% cysteine content), and high glycine/tyrosine (Powell and Rogers, 1997). According to the unified nomenclature proposed by Rogers and Powell (Rogers and Powell, 1993), these KAP are further subdivided into 23 distinct families (*KAP1.n*–*KAP23.n*) (Powell and Rogers, 1997; Rogers *et al*, 2001, 2002). In humans, five clusters of *KAP* genes, which include more than 80 individual *KAP* genes, have been

characterized (Rogers *et al*, 2001, 2002, 2004; Shibuya *et al*, 2004; Yahagi *et al*, 2004). Among them, a cluster harboring four high sulfur (*KAP1*–3 and 16) and three ultrahigh sulfur (*KAP4*, 9, and 17) *KAP* families is located on chromosome 17q12–21 (Rogers *et al*, 2001). As a rule, the *KAP1*–4 and 9 family members in this cluster contain a high degree of double cysteine-containing pentapeptide repeat structures (Rogers *et al*, 2001). Recently, we reported that two human *KAP1* genes, *KAP1.1B* and *KAP1.3*, have several size polymorphisms that are mainly due to differences in the length of the sequences encoding the cysteine-rich repeat segments (Shimomura *et al*, 2002b). This result led us to wonder whether other *KAP* families in this cluster have size polymorphisms, as well.

The *KAP* gene cluster sequence on chromosome 17q12–21 is contained in five previously sequenced human bacterial artificial chromosome (BAC) clones: AC003958, AC006070, AC025904, AC037482, and AC007455 (Rogers *et al*, 2001). Of these, AC003958, AC006070, and AC007455 are fully sequenced, and AC003958 overlaps with one end of AC006070 (Rogers *et al*, 2001). Between AC006070 and AC007455, however, there remains an incomplete region, because both AC025904 and AC037482 are partially sequenced clones that consist of several fragments (Rogers *et al*, 2001). A P1 artificial chromosome clone termed PAC10 that was isolated by Rogers *et al* (2001) overlaps with one end of AC007455, and partially covers the uncertain region between AC006070 and AC007455. On one side of PAC10, four *KAP4* genes and one *KAP4* pseudogene are located (Rogers *et al*, 2001). The *KAP4* family is one of the ultrahigh sulfur *KAP*, and a total of 13 human *KAP4* genes, *KAP4.1*–*4.12* and *KAP4.14*, have been isolated from the three BAC clones, AC006070,

Abbreviations: BAC, bacterial artificial chromosome; KAP, keratin-associated proteins; KIF, keratin intermediate filaments; ORF, open reading frame; PAC, P1 artificial chromosome

AC025904, and AC037482 (Rogers *et al*, 2001). So far, the accurate location of 11 *KAP4* genes, *KAP4.1-4.9*, *KAP4.11*, and *KAP4.12*, has been determined on chromosome 17q, whereas the location of the remaining 2 *KAP4* genes, *KAP4.10* and *KAP4.14*, has been uncertain (Rogers *et al*, 2001). Furthermore, two additional *KAP4* genes with unknown chromosomal location, *KAP4.13* and *KAP4.15*, have been identified from a human scalp cDNA library (Rogers *et al*, 2001).

In this study, using a recently fully sequenced BAC clone, AC100808, which covers the incomplete region described above, we completely characterized the domain for the *KAP4* genes on chromosome 17q, and analyzed it in detail. Our findings show that the *KAP4* family members contained abundant size polymorphisms due to variations in the repeat segments.

Results

Characterization of the BAC clone AC100808 and the identification of size polymorphisms in the *KAP4* genes

The fully sequenced BAC clone AC100808 covered about 16 kb of one end of AC006070, and corresponded to about 37 kb of one end of AC007455. Therefore, AC100808 completely connected AC006070 and AC007455 to each other (Fig 1). Analysis of the sequences of AC100808 revealed that this clone contained a group of *KAP4* genes, which was followed by groups of *KAP2* and *KAP1* genes (Fig 1). In the *KAP4* group, a total of 8 *KAP4* genes and one

KAP4 pseudogene were located on AC100808 (Fig 1). As shown in Fig 1, between the *KAP4.12* and *KAP4.9* genes on this clone, we found the *KAP4.14* gene, which had originally been identified from a partially sequenced BAC clone, AC037482 (Rogers *et al*, 2001). At the corresponding position on PAC10, however, the *KAP4.11* gene was located instead (Fig 1). Comparisons between *KAP4.11* and *KAP4.14* showed high sequence homology, with the main differences being in the length of their open reading frames (ORF), which caused differences in the number of double cysteine-containing pentapeptide repeat structures at the protein level (Fig 2). These results suggested that the two genes are actually related to each other as size polymorphisms of one gene. In addition, we found a putative novel *KAP4* gene next to the *KAP4.4* gene on AC100808, whereas the *KAP4.5* gene was located at the same position on AC006070 (Fig 1). The comparison between *KAP4.5* and this novel *KAP4* gene revealed complete nucleotide sequence identity in their ORF as well as their 5'- and 3'-non-coding regions, except for a 15-nucleotide deletion and one single-nucleotide substitution in the ORF of the novel gene, which resulted in the deletion of five amino acid residues, "CCRPS", and the substitution of one amino acid residue, "H>R", at the protein level (Fig S1a). These data suggest that the *KAP4.5*-like gene on AC100808 is a polymorphic form of the *KAP4.5* gene; we therefore named it *KAP4.5-v1* (v, variant) in this paper (Fig 1, Fig S1a). Likewise, we also identified size polymorphisms of the *KAP4.7* and *KAP4.8* genes on AC100808, which we termed *KAP4.7-v1* and *KAP4.8-v1*, respectively (Fig 1, Fig S1b,c). These polymor-

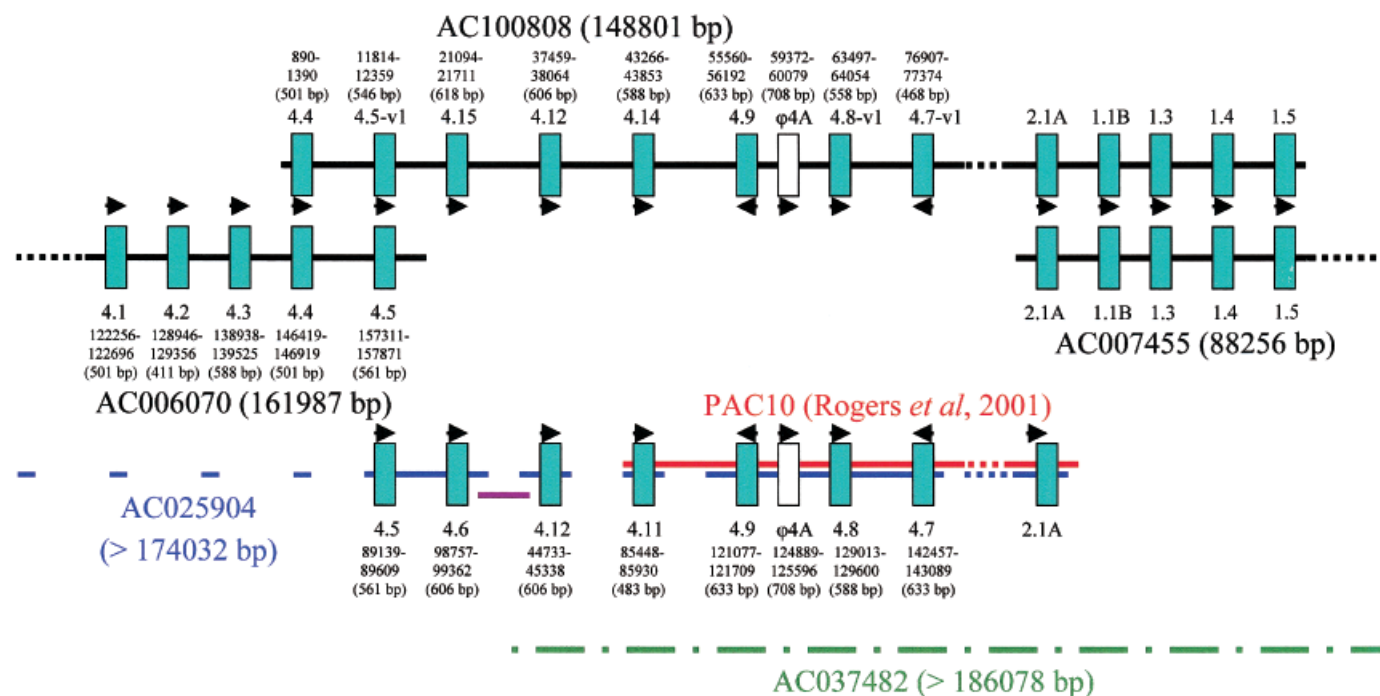


Figure 1

Physical map of the human *KAP4* genes. Black lines represent fully sequenced BAC clones, AC006070, AC100808, and AC007455. Blue and Green lines show partially sequenced BAC clones AC025904 and AC037482, respectively. The text below or above the lines indicates the BAC clone accession numbers and the size of each clone in bp. The red line refers to PAC10, isolated by Rogers *et al* (2001). Green boxes indicate the *KAP4* genes; the white box shows the *KAP4* pseudogene. The numbers just below or above the boxes refer to the names of the respective *KAP* genes. The position and size of the open reading frames of the *KAP4* genes are also indicated. Black arrowheads show the direction of gene transcription. The purple bar between *KAP4.6* and *KAP4.12* on AC025904 indicates a region closed by long-range PCR (Rogers *et al*, 2001). Note that three *KAP2* genes and one *KAP2* pseudogene, which are located between *KAP4.7* and *KAP2.1A*, are not shown.

KAP4.11	1	TTCACCTCTCCTAGAAACCAACTAGATCCTTCACCCCTGACACC
KAP4.14	1	*****
ATGGTAAACTCCTGTTGGCTCCGTGTGCTCTCACCAAGGCTGTGGCCAAGACCTCTGCCAGGAGACCTGCTGCCGCCCCAGCTGCTGT		
*****G*****		
KAP4.11	M V N S	C C G S V C S H Q G C G Q D L C Q E T C C R P S C C
KAP4.14	M V N S	C C G S V C S H Q G C G R D L C Q E T C C R P S C C
GAGACCACCTGCTGCAGGACCCTACTGTGCGCCCAAGC-----TGCTGTGTGTCCAGCTGCTGCAGGCCCAAGTGTGCCAGTCTGTGTGCTGC		
E T T C C R T T Y C R P S		
E T T C C R T T Y C R P S C C V S S C C R P Q C C Q S V C C		
-----TGCTGTGTGTCCAGCTGCTGCAAGCCCCAGTGTGCTGC		
CAGCCACCTGCTGCCGCCCCAGATGCTGCATCTCCAGTGTGCTGCGCCCAAGC-----		
C C V S S C C K P Q C C		
Q P T C C R P R C C I S S C C R P S C C V S S C C K P Q C C		
CAGTCTATGTGTGCCAGCCCACTGTGCTGCCGCCCCAGATGCTGCATCTCCAGTGTGCTGCGCCCAAGTGTGTGTGTCCAGTGTGCTGC		

Q S M C C Q P T C C R P R C C I S S C C R P S C C V S S C C		
Q S M C C Q P T C C R P R C C I S S C C R P S C C V S S C C		
AGACCCAGTGTGCCAGTTTGTGTGCTGCCAGCCCACTGTGCCAGCCCAAGTGTGCATCTCCAGTGTGCTGCCGCCCCCTCTTGTGCTGT		
*****C*****		
R P Q C C Q F V C C Q P T C C H P S C S I S S C C R P S C C		
R P Q C C Q S V C C Q P T C C H P S C S I S S C C R P S C C		
GAATCCAGTGTGCCGCCCCCTGTGCTGCTGCCAGTCTGTGCCGAGTCTCTGCCACCACTTGTATCGCCCAACCTGTGCTGC		

E S S C C R P C C C L R P V C G R V S C H T T C Y R P T C V		
E S S C C R P C C C L R P V C G R V S C H T T C Y R P T C V		
ATCTCCAGTGTGCCGCCCCCTTGTGCTGTGCCCTCTTGTGCTGAGCCCACTGCGCTGCTCATCTCTCCCTTCACTGCAAGGCCCACT		

I S S C P R P L C C A S S C C		
I S S C P R P L C C A S S C C		
AGTTGTAGACCA	582	
*****	687	

Figure 2

Nucleotide sequences of the KAP4 genes and their size polymorphisms, and the deduced amino acid sequences. KAP4.11/KAP4.14. Asterisks below the nucleotide sequence indicate sequence identity. Amino acid differences between the sequences are indicated in *bold*. The primer sequences for PCR are underlined. Dashes denote gaps in the sequence to maximize alignment. Double cysteine-containing pentapeptide repeats are boxed. The accession numbers for the KAP4 genes are given in Fig 1.

phic alleles shared almost completely identical sequences with the respective original KAP4 genes, with the exception of the deletion of sequences encoding the cysteine-rich repeat segments as well as some single-nucleotide substitutions in their ORF (Fig S1b,c). Furthermore, between the KAP4.5-v1 and KAP4.12 genes on AC100808, where KAP4.6 was located on AC025904, the KAP4.15 gene was found instead (Fig 1). KAP4.15 was originally isolated from a human scalp cDNA library and had not yet been identified at the genomic DNA level (Rogers *et al*, 2001). This cDNA clone of KAP4.15, however, lacked the sequences encoding the N-terminal and the 5'-non-coding regions (Rogers *et al*, 2001). Thus, our analysis of AC100808 finally revealed the full sequence and the chromosomal location of the KAP4.15 gene. Its sequence turned out to be completely identical to that of the KAP4.6 gene, except for a 12-nucleotide insertion into the ORF of KAP4.15, so the deduced proteins differed by a four amino acid sequence, "CCPS" (Fig S1d). Collectively, the data indicated that KAP4.15 is actually a size polymorphism of KAP4.6.

Our analysis demonstrated that the KAP gene cluster on chromosome 17q12-21 harbors 11 KAP4 genes with size polymorphisms in at least five of them (Fig 1). In addition, 13 of 15 previously reported KAP4 genes could be identified in this cluster, because KAP4.14 and KAP4.15 were shown to be size polymorphisms of KAP4.11 and KAP4.6, respectively. The remaining 2 KAP4 genes, KAP4.10 and KAP4.13, were not found in this analysis. Of these, KAP4.10 was previously identified in one fragment of a partially sequenced

BAC clone, AC025904 (Rogers *et al*, 2001). This BAC clone is now completely covered by three fully sequenced BAC clones: AC006070, AC100808, and AC007455 (Fig 1). Thus, KAP4.10 should have been isolated from these clones. None of them, however, contained the KAP4.10 gene, indicating that this gene was probably a size polymorphism of another KAP4 gene. We focused on the KAP4.1 gene, which was located on AC006070, as a candidate for this, because it shared nearly identical sequences in the 5'- and 3'-non-coding regions with KAP4.10, despite the differences in their ORF. In addition, the cDNA sequence of KAP4.13, which had previously been identified only at the mRNA level (Rogers *et al*, 2001), showed quite high homology with the genomic sequence of the KAP4.4 gene (for detailed comparisons, see below). Therefore, we speculated that KAP4.13 might be a size polymorphism of KAP4.4.

Detection of numerous size polymorphisms in the KAP4 genes by PCR To further analyze the size polymorphisms in the KAP4 genes in detail, we performed PCR amplification of the KAP4 genes using the genomic DNA of 20 Japanese individuals (nos. 1-20). It should be noted that, due to the high sequence homology between KAP4.1 and KAP4.10, common primers were used to amplify these two genes (Table I). For the same reason, we used common primers to amplify KAP4.4 and KAP4.13, as well (Table I). In addition, the primers for the KAP4.15 and KAP4.14 were common to KAP4.6 and KAP4.11, respectively, based on the evidence described above (Table I). As shown in Fig 3a,

Table I. Primer pairs for PCR amplification of the *KAP4* genes

Gene	PCR primers	Annealing temperature (°C)	Product size (bp)
<i>KAP4.1 (KAP4.10)</i>	KAP4.1&10F: 5'-catgggataaggagattccac-3' (AC006070: nt 122,179-122,200)	60	562 (<i>KAP4.1</i>)
	KAP4.1&10R: 5'-catgggatggtgactggctc-3' (AC006070: nt 122,721-122,740)		505 (<i>KAP4.10</i>)
<i>KAP4.2</i>	KAP4.2F: 5'-tctgatgacatcaggccttagc-3' (AC006070: nt 128,831-128,852)	62	622
	KAP4.2R: 5'-actgaggtgtccaattggcct-3' (AC006070: nt 129,431-129,452)		
<i>KAP4.3</i>	KAP4.3F: 5'-tcattctcctgggaaccac-3' (AC006070: nt 138,894-138,913)	58	727
	KAP4.3R: 5'-cgacatcaggaagtcacatgtt-3' (AC006070: nt 139,598-139,620)		
<i>KAP4.4 (KAP4.13)</i>	KAP4.4&13F: 5'-ctccttcctggaaaccac-3' (AC006070: nt 146,376-146,370)	62	635 (<i>KAP4.4</i>)
	KAP4.4&13R: 5'-agatactgcaggaaggctgg-3' (AC006070: nt 146,991-147,010)		455 (<i>KAP4.13</i>)
<i>KAP4.5</i>	KAP4.5F: 5'-tagaaatccaccagaaacctcc-3' (AC006070: nt 157,276-157,297)	60	676 (<i>KAP4.5</i>)
	KAP4.5R: 5'-ttccagtgtgctggctga-3' (AC006070: nt 157,932-157,951)		661 (<i>KAP4.5-v1</i>)
<i>KAP4.6 (KAP4.15)</i>	KAP4.6&15F: 5'-gcagcaaggagactccaaa-3' (AC100808: nt 21,020-21,039)	60°C	736 (<i>KAP4.6</i>)
	KAP4.6&15R: 5'-ggctgtcttctgtgaggag-3' (AC100808: nt 21,749-21,767)		748 (<i>KAP4.15</i>)
<i>KAP4.7</i>	KAP4.7F: 5'-aaacccaccagatcctcc-3' (AC100808: nt 77,388-77,406)	60	804 (<i>KAP4.7</i>)
	KAP4.7R: 5'-tggtgggctcttgaagca-3' (AC100808: nt 76,768-76,787)		639 (<i>KAP4.7-v1</i>)
<i>KAP4.8</i>	KAP4.8F: 5'-tttactctcttgaacccaatag-3' (AC100808: nt 63,452-63,476)	58	686 (<i>KAP4.8</i>)
	KAP4.8R: 5'-gtctacattgtggaccagcg-3' (AC100808: nt 64,086-64,106)		655 (<i>KAP4.8-v1</i>)
<i>KAP4.9</i>	KAP4.9F: 5'-tcactctcttgaacccacc-3' (AC100808: nt 56,216-56,236)	58	817
	KAP4.9R: 5'-gtgatgggctcattggaacatg-3' (AC100808: nt 55,420-55,442)		
<i>KAP4.11 (KAP4.14)</i>	KAP4.11&14F: 5'-ttcactctcctagaacccaac-3' (AC100808: nt 43,221-43,242)	60	582 (<i>KAP4.11</i>)
	KAP4.11&14R: 5'-tggtctacaactgtggcct-3' (AC100808: nt 43,888-43,907)		687 (<i>KAP4.14</i>)
<i>KAP4.12</i>	KAP4.12F: 5'-cagaacactcactctcctgg-3' (AC100808: nt 37,407-37,426)	58	773
	KAP4.12R: 5'-gaggtttgcttatgggaccc-3' (AC100808: nt 38,160-38,179)		

nt, nucleotide.

many size variants were detected by PCR for most of the *KAP4* genes.

The PCR using primers common to the *KAP4.1* and *KAP4.10* genes amplified a single fragment in 18 Japanese individuals, and two distinct fragments in the others (Fig 3a). Direct sequencing showed that the shorter, 505 bp fragment

was the *KAP4.10* gene, and the longer, 562 bp product corresponded to the *KAP4.1* gene. If the two genes had had independent locations, two distinct fragments should have been amplified in all the individuals. Therefore, the results of the PCR show that *KAP4.10* is a size polymorphism of *KAP4.1*, even though we did not perform pedigree analysis

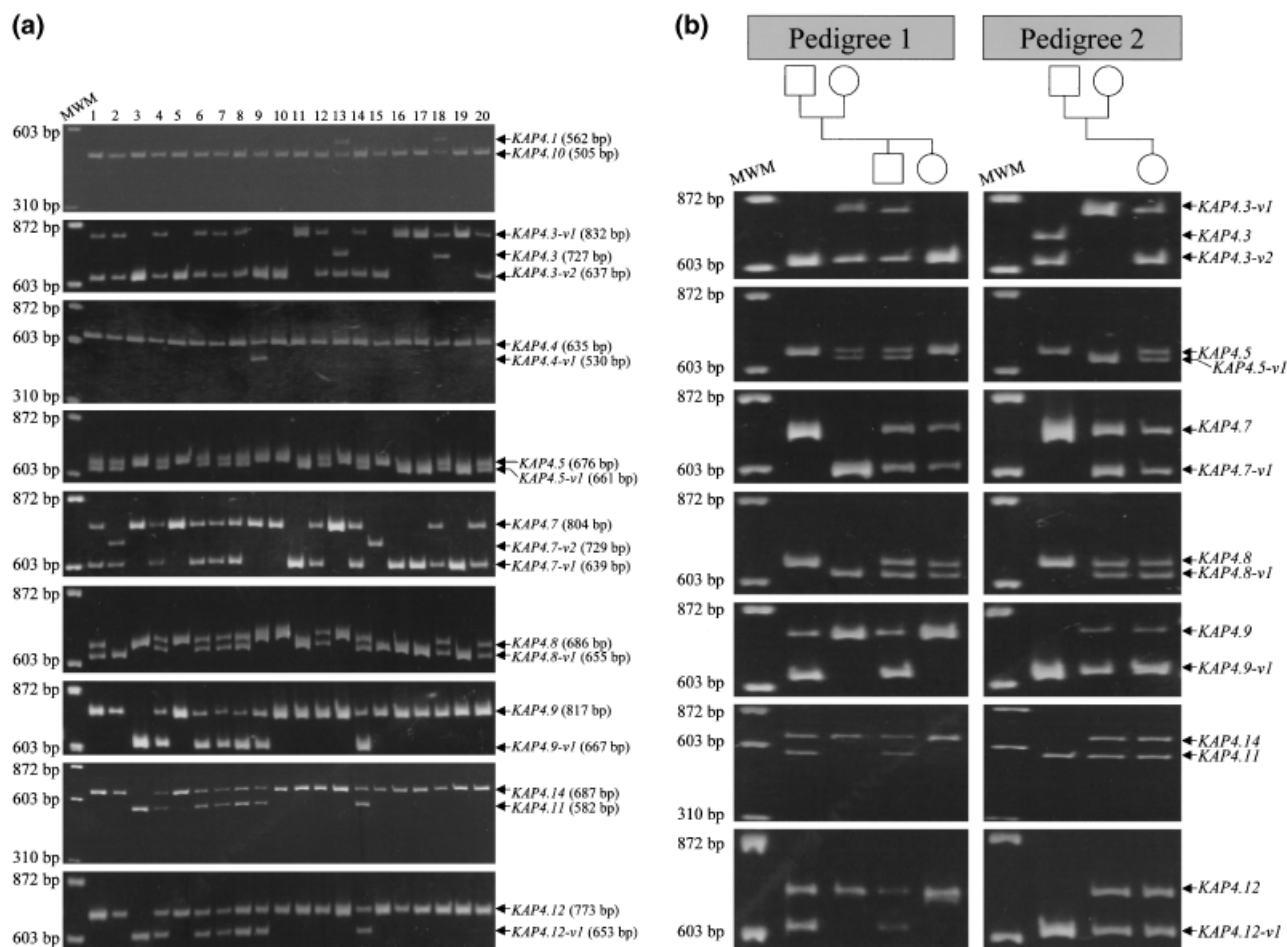


Figure 3
Detection of size polymorphisms in the *KAP4* genes by PCR. PCR amplification of the *KAP4* genes in 20 unrelated Japanese individuals (nos. 1–20) (a), and two Japanese families (b). The names of the polymorphic alleles corresponding to the amplified fragments and the size of the products are indicated on the right-hand side. MWM, molecular weight markers.

in a Japanese family with members having both alleles. The ORF of *KAP4.10* was identical to that of *KAP4.1* except for a 57 nucleotide deletion and single-nucleotide substitutions in three codons, which resulted in a 19 –amino acid residue deletion and three amino acid residue changes at the protein level (Fig S1e).

For the *KAP4.2* gene, we did not detect any size variants in this experiment (data not shown). In contrast, PCR amplification of the *KAP4.3* gene, which was located next to *KAP4.2*, led to the detection of three different products (Fig 3a). Direct sequencing revealed that one fragment, of 727 bp, corresponded to the original *KAP4.3* gene, and the others, of 832 and 627 bp, were size polymorphisms of the *KAP4.3* gene, which we termed *KAP4.3-v1* and *KAP4.3-v2*, respectively (Fig S1f). Their relationship could be confirmed by pedigree analysis (Fig 3b). As shown in Fig 2g, the differences among the *KAP4.3*, *KAP4.3-v1*, and *KAP4.3-v2* were completely due to differences in the length of sequences encoding the cysteine-rich pentapeptide repeats.

In the PCR using primers common to *KAP4.4* and *KAP4.13*, two distinct fragments were amplified in one Japanese individual, whereas only a single fragment was amplified in all the others (Fig 3a). Direct sequencing demonstrated that the product of 635 bp corresponded to

the *KAP4.4* gene, and the other, of 530 bp, was a size polymorphism of the *KAP4.4* gene, which was termed *KAP4.4-v1* (Fig S1g). When compared with the *KAP4.4* gene, *KAP4.4-v1* lacked the 105 bp nucleotide sequence that encoded the cysteine-containing repeated segments (Fig S1g). In this PCR, however, we did not detect the *KAP4.13* gene in any of the subjects. *KAP4.13* shared an identical sequence with *KAP4.4* except for a 180-bp deletion and one nucleotide substitution in *KAP4.13* (Fig S1g). Like *KAP4.4-v1*, the deletion caused differences in the length of the repeat structures at the protein level (Fig S1g). The failure of PCR to amplify *KAP4.13*, together with the high sequence homology between *KAP4.4* and *KAP4.13*, indicate that *KAP4.13* is not an independent gene, but a size polymorphism of the *KAP4.4* gene.

As shown in Fig 3a, we succeeded in amplifying the *KAP4.5-v1*, *KAP4.7-v1*, *KAP4.8-v1*, and *KAP4.14* genes, which were identified on AC100808, as size polymorphisms of the *KAP4.5*, *KAP4.7*, *KAP4.8*, and *KAP4.11* genes, respectively (Fig 1). The pedigree analysis clearly showed that these size variants segregated as a normal Mendelian trait (Fig 3b). Furthermore, we also found another size polymorphism of the *KAP4.7* gene, which we termed *KAP4.7-v2* (Fig S1b, Fig 3a).

Unexpectedly, PCR using the KAP4.6&15F and KAP4.6&15R primers resulted in amplification of only a single fragment, of 748 bp, in all the samples (data not shown). The sequence of the product corresponded to KAP4.15. As described above, KAP4.6 and KAP4.15 are related to each other as size polymorphisms (Fig 1). Thus, this PCR result showed that all the Japanese individuals analyzed here were homozygous for the KAP4.15 allele.

Finally, the PCR amplification of KAP4.9 and KAP4.12 also resulted in the identification of a size polymorphism in each gene, which we named KAP4.9-v1 and KAP4.12-v1, respectively (Fig 3a,b). KAP4.9-v1 was identical to KAP4.9, but shorter than KAP4.9 due to nucleotide sequence deletions at two distinct positions (Fig S1h). The deleted sequences, of 120 bp and 30 bp, encoded the cysteine-rich repeat structures (Fig S1h). Comparisons between KAP4.12 and KAP4.12-v1 revealed that their sequences were identical except for a 120-nucleotide deletion and single nucleotide substitutions in two codons within the ORF of KAP4.12-v1, which led to a deletion of 40 amino acid residues representing repeat segments, and two amino acid residue substitutions (Fig S1i).

Transcripts of the KAP4 gene variants are expressed in the hair follicles To demonstrate the follicular expression of the polymorphic alleles of the KAP4 genes at the mRNA level, we performed RT-PCR using total RNA from the freshly plucked hair follicles of one of the Japanese subjects (no. 6), who was confirmed to be heterozygous for KAP4.3-v1/KAP4.3-v2, KAP4.5/KAP4.5-v1, KAP4.7/KAP4.7-v1, KAP4.8/KAP4.8-v1, KAP4.9/KAP4.9-v1, KAP4.11/KAP4.14, and KAP4.12/KAP4.12-v1 at the genomic DNA level (Fig 3a). Because all the known KAP genes consist of only one exon, care was taken to eliminate traces of genomic DNA from the follicular RNA samples by digestion with DNase I. As shown in Fig 4, two well-separated cDNA were amplified

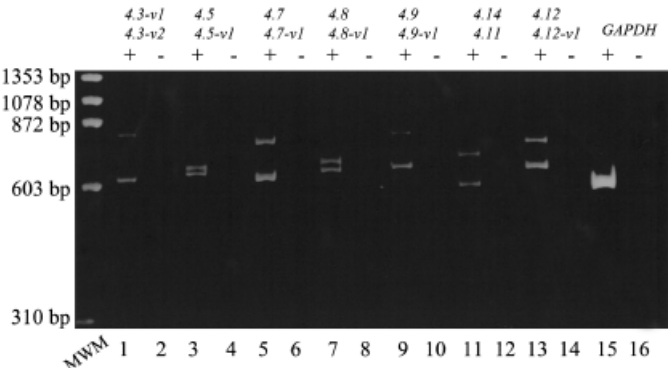


Figure 4
RT-PCR expression of the KAP4 genes and their size variants. RT-PCR amplification of the mRNAs of KAP4.3-v1/KAP4.3-v2 (lane 1), KAP4.5/KAP4.5-v1 (lane 3), KAP4.7/KAP4.7-v1 (lane 5), KAP4.8/KAP4.8-v1 (lane 7), KAP4.9/KAP4.9-v1 (lane 9), KAP4.11/KAP4.14 (lane 11), KAP4.12/KAP4.12-v1 (lane 13), and GAPDH (lane 15) from the follicular RNA of a Japanese individual (subject no. 6). GAPDH mRNA, of 597 bp, was amplified as a control, using primers located in exons 3 and 7 of the GAPDH gene (lane 15). Note that the genomic 1101 bp GAPDH fragment was clearly absent from lane 15. PCR without reverse transcription did not lead to any products (lanes 2, 4, 6, 8, 10, 12, 14, and 16), indicating that there was no genomic DNA contamination in the samples. +/– denotes reactions with or without reverse transcription. MMW, molecular weight markers.

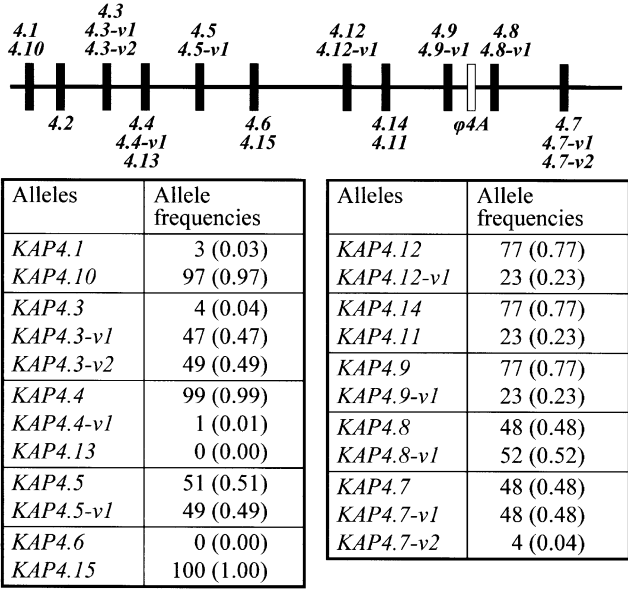


Figure 5
Allele frequencies of the KAP4 genes and their respective polymorphisms in 50 Japanese individuals. To aid understanding, a physical map of the KAP4 gene domain on chromosome 17q is shown above the tables of allele frequencies.

in the PCR for each of the KAP4 transcripts analyzed here. The sizes of the products and their sequences were completely consistent with those obtained by the PCR of genomic DNA (See the lane of individual no. 6 in Fig 3a). These results clearly showed that the transcripts of the KAP4 gene variants are expressed in the hair follicles.

Allele frequencies of the size polymorphisms in the KAP4 genes *In toto*, we performed PCR amplification of the KAP4 genes in 50 unrelated Japanese individuals (100 alleles). Based on the results, we summarized the allele frequencies of the size polymorphisms in the KAP4 genes. As shown in Fig 5, the Japanese population has the KAP4.1, KAP4.3, KAP4.4-v1, KAP4.13, KAP4.6, and KAP4.7-v2 alleles at a low frequency (0.00–0.04), and the other polymorphic alleles at relatively high frequencies (0.23–1.00). In addition, the frequencies of the KAP4.12/KAP4.12-v1, KAP4.14/KAP4.11, and KAP4.9/KAP4.9-v1 alleles were identical (Fig 5).

Discussion

We previously showed the existence of size polymorphisms in two human KAP1 genes, KAP1.1B and KAP1.3, which were mainly due to different lengths of the sequences encoding the cysteine-rich repeat structures (Shimomura *et al*, 2002b). These results led us to analyze other human KAP families. As the next step, we focused on the KAP4 family located on chromosome 17q12–21 (Rogers *et al*, 2001). The KAP4 family, the largest KAP family, is composed of 15 individual members, KAP4.1–4.15 (Rogers *et al*, 2001). By analyzing the sequences of a BAC clone, AC100808, we completely characterized the domain of the KAP4 genes on chromosome 17q, which consisted of a total of 11 KAP4 genes (Fig 1).

Furthermore, we demonstrated that four *KAP4* genes, *KAP4.1*, *KAP4.13*, *KAP4.15*, and *KAP4.14*, of which the accurate locations were previously unknown, are actually size polymorphisms of other *KAP4* genes on chromosome 17q (Fig 5). Thus, all 15 of the previously reported *KAP4* genes turned out to be located on chromosome 17q, which creates some confusion in using the present nomenclature for *KAP*. Similar complications have arisen for the human *KAP1* genes as well, because the *KAP1.1A*, *KAP1.6*, and *KAP1.7* genes, which were originally thought to be independent genes, are in actuality, size polymorphisms of the *KAP1.1B* gene, and the *KAP1.2*, *KAP1.8*, and *KAP1.9* genes are polymorphic forms of the *KAP1.3* gene (Shimomura *et al*, 2002b). These results provide the caveat that if a clone encoding a putative novel *KAP* gene is isolated, there is a chance that it is a size polymorphism of another, previously reported *KAP* gene. Therefore, in this paper, we termed the additional size polymorphisms of the *KAP4* genes as "*KAP4.m-vn*" so as not to cause further nomenclature problems.¹

In this study, we identified size polymorphisms in 10 of 11 *KAP4* genes on chromosome 17q (Fig 5). The deduced amino acid sequence alignment showed that all the *KAP4* proteins contained high numbers of cysteine-rich pentapeptide repeats (Rogers *et al*, 2001), and the differences in size of most of the *KAP4* gene variants were completely due to changes in the length of the sequences encoding these repeat structures (Fig 2 and S1a–i). These results indicate that the size polymorphisms in the *KAP4* genes may have arisen by intragenic deletion and/or duplication events of the repeated segments during evolution.

In general, most size variants of the *KAP4* genes identified in this study are deletion polymorphisms. It is surprising that some of them lose almost half of the length of the central repeat domain (Fig 2 and S1a–i). The mechanism of how such large deletion events occurred in the *KAP4* genes remains unknown. Previous studies have shown that several human keratin genes have deletion polymorphisms due to variations in the sequences encoding a particular structural protein motif termed the glycine loop that contains glycine-rich repeats (Korge *et al*, 1992a,b; Wanner *et al*, 1993). Wanner *et al* suggested that slipped-strand mispairing followed by illegal crossing-over would account for the interstitial deletions in the keratin genes (Wanner *et al*, 1993). There is a possibility that, in part, the same mechanism may have been involved in generating the deletion polymorphisms of the *KAP4* genes.

The allele frequencies of the *KAP4* gene variants showed that the Japanese population has both the *KAP4.1* and *KAP4.3* alleles at a low frequency (Fig 5). In addition, neither the *KAP4.13* nor the *KAP4.6* allele was detected in the Japanese population (Fig 5). These polymorphic alleles were originally isolated from genomic DNA and/or a cDNA library derived from other human populations (Rogers *et al*, 2001). We previously showed the existence of population-specific polymorphisms in the human *KAP1* genes (Shimomura *et al*, 2002b). It is possible that several size polymorphisms in the

KAP4 genes also exhibit population-specificity. Furthermore, the allele frequencies of the *KAP4* gene variants identified in this study may show variation among different populations. We plan to analyze the *KAP4* genes in other human populations in our future work, which may increase our understanding of how the human *KAP4* genes evolved.

In addition to the results mentioned above, we obtained some other interesting observations in this study. As shown in Fig 3a, the fragment patterns of *KAP4.9/KAP4.9-v1*, *KAP4.14/KAP4.11*, and *KAP4.12/KAP4.12-v1* were virtually identical. The same observation was obtained in 30 additional Japanese individuals (data not shown). These results strongly indicate that the *KAP4.9*, *KAP4.14*, and *KAP4.12* alleles might be linked with each other. Linkage among the *KAP4.9-v1*, *KAP4.11*, and *KAP4.12-v1* alleles was also suggested. Likewise, comparison of the fragment patterns between the *KAP4.7* and *KAP4.8* genes led us to ascertain that the polymorphic alleles of these two genes might be linked, i.e., the *KAP4.7* allele seems to link with the *KAP4.8* allele, and both the *KAP4.7-v1* and *KAP4.7-v2* alleles are likely to link with the *KAP4.8-v2* allele (Fig 3a).

Previously performed *in situ* hybridization studies demonstrated that the transcripts of the human *KAP1* and *KAP4* genes are expressed predominantly in the highly differentiated portions of the middle and upper cortex (Rogers *et al*, 2001; Shimomura *et al*, 2002a, b, 2003). These results suggested that both *KAP1* and *KAP4* members are heavily involved in the terminal keratinization of the hair fiber cortex, probably through extensive cross-linking with the KIF via disulfide bonds. In this study, we have clearly shown, at both the genomic DNA and mRNA levels, that the human *KAP4* genes had numerous size polymorphisms due to variations in the length of the sequences encoding the cysteine-containing repeat segments. To our regret, however, we have not yet detected the size variations at the protein level, mainly because of a lack of antibodies against these proteins. It is difficult to raise antibodies against the *KAP4* proteins, due to their high cysteine content. Nevertheless, the size polymorphisms of the *KAP4* genes should result in different levels of cysteine content at the protein level. In the hair fiber, it has been proposed that the high cysteine content in *KAP* provides the chemical means for the cross-linked network with the KIF (Powell and Rogers, 1986). Thus, the different numbers of cysteine-rich repeats among the size variants in the *KAP* will change the strength of their interaction with the KIF. In addition, there is a strong possibility that the other human *KAP* family members also have size polymorphisms, and the patterns of all the *KAP* will play an important role in determining the characteristics of the hair fiber among individuals as well as among different populations. The size polymorphisms of the *KAP* genes might thus become a useful forensic tool, because a partial description of an individual could be obtained from a small DNA sample, with a high probability of accuracy. Therefore, the discovery of the complete set of size polymorphisms among the human *KAP* genes could be of great benefit in the future.

Materials and Methods

Sequence analysis of BAC clone AC100808 Homology searching of the GenBank database using a partially sequenced BAC

¹The gene symbol of "keratin-associated protein 4-m" that has been approved by the Human Genome Organization Gene Nomenclature Committee (<http://www.gene.ucl.ac.uk/nomenclature/>) is "KRTAP4-m", whose alias is "KAP4.m". Here, we proposed here a system for nomenclature of the size variants of the *KAP* genes.

clone, AC025904, led to the identification of a fully sequenced BAC clone, AC100808, which was reported recently. AC100808 has 148,801 bp, and shares identical sequences with several fragments of AC025904. The sequences of AC100808 were then used to search for DNA homology with the BAC clones AC006070, AC037482, and AC007455, as well as with the *KAP* genes registered in the GenBank database, using the BLASTN program.

Source of DNA After written informed consent was obtained, peripheral leukocyte DNA was prepared from unrelated healthy Japanese individuals. Genomic DNA was also extracted from two generations of Japanese family members. The study was conducted in accordance with the Declaration of Helsinki guidelines and was approved by the ethics commission at Niigata University.

PCR amplification of the *KAP4* genes Using the genomic DNA of Japanese individuals as templates, the *KAP4* genes were amplified by PCR. The primer pairs used are indicated in Table I. PCR was performed using Advantage 2 DNA polymerase (Clontech, Tokyo, Japan). The amplification conditions were 96°C for 3 min, followed by 30 cycles of 96°C for 30 s, the indicated annealing temperature for 30 s, and 72°C for 1 min, with a final extension at 72°C for 7 min. The amplified PCR fragments were analyzed on 6% polyacrylamide gels. After gel extraction of the fragments, direct fluorescent chain-termination DNA cycle sequencing was performed (Big Dye DNA sequencing kit, Applied Biosystems, Foster City, California). The DNA sequences were analyzed on an ABI373 DNA sequencer (Applied Biosystems).

Reverse transcriptase, RT-PCR Total RNA was isolated from 15 freshly plucked anagen hair follicles of one of the Japanese subjects (no. 6) using the Isogen Kit (Nippongene, Tokyo, Japan), according to the manufacturer's recommendations. The RNA was digested for 10 min with RNase-free DNase (Roche, Mannheim, Germany) to remove contaminating genomic DNA, and reverse transcribed using a random primer and the SuperScript II reverse transcriptase (Invitrogen, Carlsbad, CA). The cDNA were amplified by PCR using specific primers for *KAP4.3*, *KAP4.5*, *KAP4.7*, *KAP4.8*, *KAP4.9*, *KAP4.11*, and *KAP4.12* (Table I). The primers and amplification conditions were the same as those used for the PCR of genomic DNA (Table I). The PCR products were sequenced as described above.

Supplementary Material

The following material is available from <http://www.blackwellpublishing.com/products/journals/suppmat/JID/JID23662/JID23662sm.htm>

Figure S1

Nucleotide sequences of the *KAP4* genes and their size polymorphisms, and the deduced amino acid sequences.

DOI: 10.1111/j.0022-202X.2005.23662.x

Manuscript received September 25, 2004; revised November 12, 2004; accepted for publication November 30, 2004

Address correspondence to: Yutaka Shimomura, Department of Dermatology, Niigata University School of Medicine, Asahimachi-dori, Niigata 951-8510, Japan. Email: yshimo@med.niigata-u.ac.jp

References

- Korge BP, Compton JG, Steinert PM, Mischke D: The two size alleles of human keratin 1 are due to a deletion in the glycine-rich carboxyl-terminal V2-subdomain. *J Invest Dermatol* 99:697–702, 1992a
- Korge BP, Gan SQ, McBride OW, Mischke D, Steinert PM: Extensive size polymorphism of the human keratin 10 chain resides in the C-terminal V2 subdomain due to variable numbers and sizes of glycine loops. *Proc Natl Acad Sci USA* 89:910–914, 1992b
- Powell BC, Rogers GE: Hair keratin: Composition, structure and biogenesis. In: Bereiter-Hahn J, Matoltsy AG, Richards KS (eds). *Biology of the Integument*. Springer-Verlag, Berlin: 1986; p 696–721
- Powell BC, Rogers GE: The role of keratin proteins and their genes in the growth, structure and properties of hair. In: Jolles P, Zahn H, Höcker H (eds). *Formation and Structure of Human Hair*. Basel: Birkh, 1997; p 59–148
- Rogers GE, Powell BC: Organization and expression of hair follicle genes. *J Invest Dermatol* 101:50s–55s, 1993
- Rogers MA, Langbein L, Winter H, Ehmann C, Praetzel S, Korn B, Schweizer J: Characterization of a cluster of human high/ultrahigh sulfur keratin-associated protein genes embedded in the type I keratin gene domain on chromosome 17q12–21. *J Biol Chem* 276:19440–19451, 2001
- Rogers MA, Langbein L, Winter H, Ehmann C, Praetzel S, Schweizer J: Characterization of a first domain of human high glycine-tyrosine and high sulfur keratin-associated protein (KAP) genes on chromosome 21q22.1. *J Biol Chem* 277:48993–49002, 2002
- Rogers MA, Langbein L, Winter H, Beckmann I, Praetzel S, Schweizer J: Hair keratin associated proteins: Characterization of a second high sulfur KAP gene domain on human chromosome 21. *J Invest Dermatol* 122:147–158, 2004
- Shibuya K, Obayashi I, Asakawa S, Minoshima S, Kudoh J, Shimizu N: A cluster of 21 keratin-associated protein genes within introns of another gene on human chromosome 21q22.3. *Genomics* 83:679–693, 2004
- Shimomura Y, Aoki N, Rogers MA, Langbein L, Schweizer J, Ito M: hKAP1.6 and hKAP1.7, two novel human high sulfur keratin-associated proteins are expressed in the hair follicle cortex. *J Invest Dermatol* 118:226–231, 2002a
- Shimomura Y, Aoki N, Rogers MA, Langbein L, Schweizer J, Ito M: Characterization of human keratin-associated protein 1 family members. *J Invest Dermatol Symp Proc* 8:96–99, 2003
- Shimomura Y, Aoki N, Schweizer J, Langbein L, Rogers MA, Winter H, Ito M: Polymorphisms in the human high sulfur hair keratin-associated protein 1, KAP1, gene family. *J Biol Chem* 277:45493–45501, 2002b
- Wanner R, Förster HH, Tilmans I, Mischke D: Allelic variations of human keratins K4 and K5 provide polymorphic markers within the type II keratin gene cluster on chromosome 12. *J Invest Dermatol* 100:735–741, 1993
- Yahagi S, Shibuya K, Obayashi I, Masaki H, Kurata Y, Kudoh J, Shimizu N: Identification of two novel clusters of ultrahigh-sulfur keratin-associated protein genes on human chromosome 11. *Biochem Biophys Res Commun* 318:655–664, 2004